EXPERIMENT 01

Basic Feature Engineering Techniques

Python implementation using the Titanic dataset

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# Experiment 01: Basic Feature Engineering Techniques

# 1. Import libraries
import pandas as pd
import numpy as np
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler

# 2. Load Titanic dataset
df = sns.load_dataset('titanic')

# 3. Display first five records
print("First Five Records:")
print(df.head())

# 4. Check dataset information
print("\nDataset Information:")
print(df.info())

# 5. Check missing values
print("\nMissing Values Before Cleaning:")
print(df.isnull().sum())

# 6. Handle missing numerical values using Median
df['age'] = df['age'].fillna(df['age'].median())
df['fare'] = df['fare'].fillna(df['fare'].median())

# 7. Handle missing categorical values using Mode
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])

# 8. Label Encoding for Sex
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])

# 9. One-Hot Encoding for Embarked
df = pd.get_dummies(df, columns=['embarked'], dtype=int)

# 10. Feature Scaling using StandardScaler
scaler = StandardScaler()

columns = ['age', 'sibsp', 'parch', 'fare']
df[columns] = scaler.fit_transform(df[columns])

# 11. Display final dataset
print("\nFinal Clean Dataset:")
print(df.head())

# 12. Check missing values after cleaning
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

# 13. Display dataset shape
print("\nDataset Shape:")
print(df.shape)

print("\nFeature Engineering Completed Successfully!")
```